

Aman

SUMMARY

Career roadmap (2026–2028) focused on becoming an Embedded Linux Engineer, System Software Engineer, or Industrial IoT Engineer. Strongest market position identified as Linux + Python + Networking + Embedded Systems + Industrial Domain Knowledge.

EDUCATION

Diploma in Computer Science

Completed

B.Tech CSE

Currently pursuing (6th Semester completed)

EXPERIENCE

Assistant Engineer (R&D)

Joined company: 05 December 2022 – Current date: 08 June 2026

Total Experience: Approximately 3.5 years

PROJECTS

Linux Monitoring Tool

- CPU Usage
- Memory Usage
- Disk Usage
- Network Status

Railway Device Dashboard

- Python
- PostgreSQL
- React

Inventory Management System

- FastAPI
- PostgreSQL

Dockerized Application | Deploy all projects using Docker.

TECHNICAL SKILLS

Linux

Basic Python

Electron

React.js

Node.js

Networking

Railway Domain

Industrial Product Development

CURRENT SALARY

₹35,000–₹40,000 per month; Annual CTC approximately ₹4.2–₹4.8 LPA

CAREER GOAL

Become an: 1) Embedded Linux Engineer 2) System Software Engineer 3) Industrial IoT Engineer. Target Salary: 2027: ₹8–12 LPA; 2028: ₹12–20+ LPA; 2030+: ₹20–35+ LPA.

PHASE 1 (JUNE 2026 – DECEMBER 2026)

Primary Objective: Build strong fundamentals and complete B.Tech. Linux Learning Path topics: Linux File System, File Permissions, Users and Groups, Process Management, Service Management, SSH, Cron Jobs, Log Files. Commands: ps, top, htop, grep, awk, sed, systemctl, journalctl. Practice: Create backup scripts, monitoring scripts, automate repetitive tasks. Python Month 1 topics: Variables, Data Types, Conditions, Loops, Functions; practice: Calculator, Number Programs, File Operations. Month 2 topics: OOP, Classes, Objects, Inheritance, Exception Handling; practice: Student Management System, Employee Management System. Month 3 topics: JSON, CSV, Logging, Modules; projects: Log Parser, CSV Analyzer. Month 4 topics: SQLite, Database Connectivity; projects: Inventory Management, Device Record Management. Networking: OSI Model, TCP/IP, UDP, DHCP, DNS, Routing, NAT, VLAN. Tools: Wireshark, tcpdump, netstat, ss. Git: Clone, Commit, Branching, Merge, Rebase, Cherry-pick, Conflict Resolution. Practice: Use Git for every project.

PHASE 2 (JANUARY 2027 – DECEMBER 2027)

Objective: Become Linux + Python Professional Engineer. Docker topics: Containers, Images, Dockerfile, Volumes, Networks, Docker Compose; projects: Deploy Python Applications, Deploy Monitoring Dashboard. PostgreSQL topics: CRUD, Relationships, Joins, Indexes, Transactions; projects: Device Database, Employee Database. Advanced Python topics: Multithreading, Async Programming, REST APIs, Flask, FastAPI; projects: Device Monitoring Server, REST API Backend, Network Monitoring Tool. Build Portfolio includes Linux Monitoring Tool, Railway Device Dashboard, Inventory Management System, Dockerized Application.

PHASE 3 (JANUARY 2028 – DECEMBER 2028)

Objective: Move into Embedded Linux and System Engineering. C Programming topics: Variables, Functions, Arrays, Pointers, Structures, Dynamic Memory Allocation; projects: File Utilities, System Utilities. Socket Programming topics: TCP Client, TCP Server, UDP Communication; projects: Chat Application, File Transfer System. Embedded Linux topics: Linux Boot Process, U-Boot, Device Tree, Buildroot, Yocto, Cross Compilation; projects: Custom Linux Build, Embedded Monitoring Tool. Linux Internals: Process Scheduling, Memory Management, Threads, IPC. Industrial Protocols: MQTT, Modbus, RS232, RS485, CAN Bus.

DAILY STUDY PLAN

Monday to Friday: Python 1 Hour, Linux 30 Minutes, Networking 30 Minutes (Total 2 Hours). Saturday: Project Work 3–4 Hours. Sunday: Revision 2 Hours, GitHub Updates 1 Hour, Documentation 1 Hour.

RESUME TARGET BY END OF 2027

Skills: Linux, Python, Docker, PostgreSQL, Git, Networking. Projects: Linux Monitoring Tool, REST API Server, Inventory Management System, Railway Device Dashboard. Experience: 4.5–5 Years. Expected Salary: ₹8–12 LPA.

RESUME TARGET BY END OF 2028

Skills: Linux, Python, C, Embedded Linux, Docker, Networking, MQTT. Experience: 5.5–6 Years. Expected Salary: ₹12–20+ LPA.

FINAL RECOMMENDATION

Do NOT focus primarily on becoming a React Developer. Strongest market position: Linux + Python + Networking + Embedded Systems + Industrial Domain Knowledge. Focus on becoming a Linux/System/Embedded Engineer rather than a general web developer.